

CHAPTER 14—SIMPLE LINEAR REGRESSION

MULTIPLE CHOICE

1. In a regression analysis, the error term ε is a random variable with a mean or expected value of
 - a. zero
 - b. one
 - c. any positive value
 - d. any value
2. The coefficient of determination
 - a. cannot be negative
 - b. is the square root of the coefficient of correlation
 - c. is the same as the coefficient of correlation
 - d. can be negative or positive
3. If the coefficient of determination is a positive value, then the coefficient of correlation
 - a. must also be positive
 - b. must be zero
 - c. can be either negative or positive
 - d. must be larger than 1
4. In regression analysis, the model in the form $y = \beta_0 + \beta_1 x + \varepsilon$ is called
 - a. regression equation
 - b. correlation equation
 - c. estimated regression equation
 - d. regression model
5. The mathematical equation relating the independent variable to the expected value of the dependent variable; that is, $E(y) = \beta_0 + \beta_1 x$, is known as
 - a. regression equation
 - b. correlation equation
 - c. estimated regression equation
 - d. regression model
6. The model developed from sample data that has the form of $\hat{y} = b_0 + b_1 x$ is known as
 - a. regression equation
 - b. correlation equation
 - c. estimated regression equation
 - d. regression model
7. In the following estimated regression equation $\hat{y} = b_0 + b_1 x$
 - a. b_1 is the slope
 - b. b_1 is the intercept
 - c. b_0 is the slope
 - d. None of these alternatives is correct.
8. In regression analysis, the unbiased estimate of the variance is
 - a. coefficient of correlation
 - b. coefficient of determination
 - c. mean square error
 - d. slope of the regression equation

9. The interval estimate of the mean value of y for a given value of x is
 - a. prediction interval estimate
 - b. confidence interval estimate
 - c. average regression
 - d. x versus y correlation interval
10. The interval estimate of an individual value of y for a given value of x is
 - a. prediction interval estimate
 - b. confidence interval estimate
 - c. average regression
 - d. x versus y correlation interval
11. The standard error is the
 - a. t-statistic squared
 - b. square root of SSE
 - c. square root of SST
 - d. square root of MSE
12. If only MSE is known, you can compute the
 - a. r square
 - b. coefficient of determination
 - c. standard error
 - d. all of these alternatives are correct
13. The value of the coefficient of correlation (R)
 - a. can be equal to the value of the coefficient of determination (R^2)
 - b. can never be equal to the value of the coefficient of determination (R^2)
 - c. is always smaller than the value of the coefficient of determination
 - d. is always larger than the value of the coefficient of determination
14. In a regression analysis the standard error is determined to be 4. In this situation the MSE
 - a. is 2
 - b. is 16
 - c. depends on the sample size
 - d. depends on the degrees of freedom
15. In regression analysis, which of the following is not a required assumption about the error term ϵ ?
 - a. The expected value of the error term is one.
 - b. The variance of the error term is the same for all values of X.
 - c. The values of the error term are independent.
 - d. The error term is normally distributed.
16. A regression analysis between sales (Y in \$1000) and advertising (X in dollars) resulted in the following equation

$$\hat{Y} = 30,000 + 4 X$$

The above equation implies that an

 - a. increase of \$4 in advertising is associated with an increase of \$4,000 in sales
 - b. increase of \$1 in advertising is associated with an increase of \$4 in sales
 - c. increase of \$1 in advertising is associated with an increase of \$34,000 in sales
 - d. increase of \$1 in advertising is associated with an increase of \$4,000 in sales
17. Regression analysis is a statistical procedure for developing a mathematical equation that describes how
 - a. one independent and one or more dependent variables are related

- b. several independent and several dependent variables are related
 - c. one dependent and one or more independent variables are related
 - d. None of these alternatives is correct.
18. In a simple regression analysis (where Y is a dependent and X an independent variable), if the Y intercept is positive, then
- a. there is a positive correlation between X and Y
 - b. if X is increased, Y must also increase
 - c. if Y is increased, X must also increase
 - d. None of these alternatives is correct.
19. In regression analysis, the variable that is being predicted is the
- a. dependent variable
 - b. independent variable
 - c. intervening variable
 - d. is usually x
20. The equation that describes how the dependent variable (y) is related to the independent variable (x) is called
- a. the correlation model
 - b. the regression model
 - c. correlation analysis
 - d. None of these alternatives is correct.
21. In regression analysis, the independent variable is
- a. used to predict other independent variables
 - b. used to predict the dependent variable
 - c. called the intervening variable
 - d. the variable that is being predicted
22. Larger values of r^2 imply that the observations are more closely grouped about the
- a. average value of the independent variables
 - b. average value of the dependent variable
 - c. least squares line
 - d. origin
23. In a regression analysis, the coefficient of determination is 0.4225. The coefficient of correlation in this situation is
- a. 0.65
 - b. 0.1785
 - c. any positive value
 - d. any value
24. In a regression analysis, the coefficient of correlation is 0.16. The coefficient of determination in this situation is
- a. 0.4000
 - b. 0.0256
 - c. 4
 - d. 2.56
25. In simple linear regression analysis, which of the following is not true?
- a. The F test and the t test yield the same conclusion.
 - b. The F test and the t test may or may not yield the same conclusion.
 - c. The relationship between X and Y is represented by means of a straight line.
 - d. The value of $F = t^2$.

26. Correlation analysis is used to determine
- the equation of the regression line
 - the strength of the relationship between the dependent and the independent variables
 - a specific value of the dependent variable for a given value of the independent variable
 - None of these alternatives is correct.
27. In a regression and correlation analysis if $r^2 = 1$, then
- SSE must also be equal to one
 - SSE must be equal to zero
 - SSE can be any positive value
 - SSE must be negative
28. In a regression and correlation analysis if $r^2 = 1$, then
- SSE = SST
 - SSE = 1
 - SSR = SSE
 - SSR = SST
29. In a regression analysis, the regression equation is given by $y = 12 - 6x$. If $SSE = 510$ and $SST = 1000$, then the coefficient of correlation is
- 0.7
 - +0.7
 - 0.49
 - 0.49
30. In a regression analysis if $SSE = 200$ and $SSR = 300$, then the coefficient of determination is
- 0.6667
 - 0.6000
 - 0.4000
 - 1.5000
31. If the coefficient of determination is equal to 1, then the coefficient of correlation
- must also be equal to 1
 - can be either -1 or +1
 - can be any value between -1 to +1
 - must be -1
32. In a regression analysis, the variable that is being predicted
- must have the same units as the variable doing the predicting
 - is the independent variable
 - is the dependent variable
 - usually is denoted by x
33. Regression analysis was applied between demand for a product (Y) and the price of the product (X), and the following estimated regression equation was obtained.

$$\hat{Y} = 120 - 10X$$

Based on the above estimated regression equation, if price is increased by 2 units, then demand is expected to

- increase by 120 units
- increase by 100 units
- increase by 20 units
- decrease by 20 units

34. The coefficient of correlation
 - a. is the square of the coefficient of determination
 - b. is the square root of the coefficient of determination
 - c. is the same as r-square
 - d. can never be negative
35. If the coefficient of correlation is a positive value, then the regression equation
 - a. must have a positive slope
 - b. must have a negative slope
 - c. could have either a positive or a negative slope
 - d. must have a positive y intercept
36. If the coefficient of correlation is 0.8, the percentage of variation in the dependent variable explained by the variation in the independent variable is
 - a. 0.80%
 - b. 80%
 - c. 0.64%
 - d. 64%
37. In regression and correlation analysis, if SSE and SST are known, then with this information the
 - a. coefficient of determination can be computed
 - b. slope of the line can be computed
 - c. Y intercept can be computed
 - d. x intercept can be computed
38. In regression analysis, if the independent variable is measured in pounds, the dependent variable
 - a. must also be in pounds
 - b. must be in some unit of weight
 - c. cannot be in pounds
 - d. can be any units
39. If there is a very weak correlation between two variables, then the coefficient of determination must be
 - a. much larger than 1, if the correlation is positive
 - b. much smaller than -1, if the correlation is negative
 - c. much larger than one
 - d. None of these alternatives is correct.
40. SSE can never be
 - a. larger than SST
 - b. smaller than SST
 - c. equal to 1
 - d. equal to zero
41. If the coefficient of correlation is -0.4, then the slope of the regression line
 - a. must also be -0.4
 - b. can be either negative or positive
 - c. must be negative
 - d. must be 0.16
42. If the coefficient of correlation is a negative value, then the coefficient of determination
 - a. must also be negative
 - b. must be zero
 - c. can be either negative or positive
 - d. must be positive

43. It is possible for the coefficient of determination to be
 - a. larger than 1
 - b. less than one
 - c. less than -1
 - d. None of these alternatives is correct.
44. If two variables, x and y, have a strong linear relationship, then
 - a. there may or may not be any causal relationship between x and y
 - b. x causes y to happen
 - c. y causes x to happen
 - d. None of these alternatives is correct.
45. If the coefficient of determination is 0.81, the coefficient of correlation
 - a. is 0.6561
 - b. could be either + 0.9 or - 0.9
 - c. must be positive
 - d. must be negative
46. A least squares regression line
 - a. may be used to predict a value of y if the corresponding x value is given
 - b. implies a cause-effect relationship between x and y
 - c. can only be determined if a good linear relationship exists between x and y
 - d. None of these alternatives is correct.
47. If all the points of a scatter diagram lie on the least squares regression line, then the coefficient of determination for these variables based on these data is
 - a. 0
 - b. 1
 - c. either 1 or -1, depending upon whether the relationship is positive or negative
 - d. could be any value between -1 and 1
48. If a data set has SSR = 400 and SSE = 100, then the coefficient of determination is
 - a. 0.10
 - b. 0.25
 - c. 0.40
 - d. 0.80
49. Compared to the confidence interval estimate for a particular value of y (in a linear regression model), the interval estimate for an average value of y will be
 - a. narrower
 - b. wider
 - c. the same
 - d. None of these alternatives is correct.
50. A regression analysis between sales (in \$1000) and price (in dollars) resulted in the following equation

$$\hat{Y} = 60 - 8X$$

The above equation implies that an

- a. increase of \$1 in price is associated with a decrease of \$8 in sales
- b. increase of \$8 in price is associated with an decrease of \$52,000 in sales
- c. increase of \$1 in price is associated with a decrease of \$52 in sales
- d. increase of \$1 in price is associated with a decrease of \$8000 in sales

51. In a regression analysis if $SST = 500$ and $SSE = 300$, then the coefficient of determination is
- 0.20
 - 1.67
 - 0.60
 - 0.40
52. Regression analysis was applied between sales (in \$1000) and advertising (in \$100) and the following regression function was obtained.

$$\hat{Y} = 500 + 4X$$

Based on the above estimated regression line if advertising is \$10,000, then the point estimate for sales (in dollars) is

- \$900
 - \$900,000
 - \$40,500
 - \$505,000
53. The coefficient of correlation
- is the square of the coefficient of determination
 - is the square root of the coefficient of determination
 - is the same as r-square
 - can never be negative
54. If the coefficient of correlation is 0.4, the percentage of variation in the dependent variable explained by the variation in the independent variable
- is 40%
 - is 16%.
 - is 4%
 - can be any positive value
55. In regression analysis if the dependent variable is measured in dollars, the independent variable
- must also be in dollars
 - must be in some units of currency
 - can be any units
 - cannot be in dollars
56. If there is a very strong correlation between two variables then the coefficient of determination must be
- much larger than 1, if the correlation is positive
 - much smaller than -1, if the correlation is negative
 - any value larger than 1
 - None of these alternatives is correct.
57. If the coefficient of correlation is 0.90, then the coefficient of determination
- is also 0.9
 - is either 0.81 or -0.81
 - can be either negative or positive
 - must be 0.81
58. A regression analysis between demand (Y in 1000 units) and price (X in dollars) resulted in the following equation

$$\hat{Y} = 9 - 3X$$

The above equation implies that if the price is increased by \$1, the demand is expected to

- a. increase by 6 units
 - b. decrease by 3 units
 - c. decrease by 6,000 units
 - d. decrease by 3,000 units
59. In a regression analysis if $SST = 4500$ and $SSE = 1575$, then the coefficient of determination is
- a. 0.35
 - b. 0.65
 - c. 2.85
 - d. 0.45
60. Regression analysis was applied between sales (in \$10,000) and advertising (in \$100) and the following regression function was obtained.

$$\hat{Y} = 50 + 8 X$$

Based on the above estimated regression line if advertising is \$1,000, then the point estimate for sales (in dollars) is

- a. \$8,050
 - b. \$130
 - c. \$130,000
 - d. \$1,300,000
61. If the coefficient of correlation is a positive value, then
- a. the intercept must also be positive
 - b. the coefficient of determination can be either negative or positive, depending on the value of the slope
 - c. the regression equation could have either a positive or a negative slope
 - d. the slope of the line must be positive
62. If the coefficient of determination is 0.9, the percentage of variation in the dependent variable explained by the variation in the independent variable
- a. is 0.90%
 - b. is 90%.
 - c. is 81%
 - d. 0.81%
63. Regression analysis was applied between sales (Y in \$1,000) and advertising (X in \$100), and the following estimated regression equation was obtained.

$$\hat{Y} = 80 + 6.2 X$$

Based on the above estimated regression line, if advertising is \$10,000, then the point estimate for sales (in dollars) is

- a. \$62,080
- b. \$142,000
- c. \$700
- d. \$700,000

NARRBEGIN: Exhibit 12-01

Exhibit 14-1

The following information regarding a dependent variable (Y) and an independent variable (X) is provided.

Y	X
4	2
3	1
4	4
6	3
8	5

SSE = 6

SST = 16

NARREND

64. Refer to Exhibit 14-1. The least squares estimate of the Y intercept is
 - a. 1
 - b. 2
 - c. 3
 - d. 4
65. Refer to Exhibit 14-1. The least squares estimate of the slope is
 - a. 1
 - b. 2
 - c. 3
 - d. 4
66. Refer to Exhibit 14-1. The coefficient of determination is
 - a. 0.7096
 - b. - 0.7906
 - c. 0.625
 - d. 0.375
67. Refer to Exhibit 14-1. The coefficient of correlation is
 - a. 0.7906
 - b. - 0.7906
 - c. 0.625
 - d. 0.375
68. Refer to Exhibit 14-1. The MSE is
 - a. 1
 - b. 2
 - c. 3
 - d. 4

NARRBEGIN: Exhibit 12-02

Exhibit 14-2

You are given the following information about y and x.

y Dependent Variable	x Independent Variable
5	1
4	2
3	3
2	4
1	5

NARREND

69. Refer to Exhibit 14-2. The least squares estimate of b_1 (slope) equals
- 1
 - 1
 - 6
 - 5
70. Refer to Exhibit 14-2. The least squares estimate of b_0 (intercept) equals
- 1
 - 1
 - 6
 - 5
71. Refer to Exhibit 14-2. The point estimate of y when $x = 10$ is
- 10
 - 10
 - 4
 - 4
72. Refer to Exhibit 14-2. The sample correlation coefficient equals
- 0
 - +1
 - 1
 - 0.5
73. Refer to Exhibit 14-2. The coefficient of determination equals
- 0
 - 1
 - +1
 - 0.5

NARRBEGIN: Exhibit 12-03

Exhibit 14-3

You are given the following information about y and x .

y Dependent Variable	x Independent Variable
12	4
3	6
7	2
6	4

NARREND

74. Refer to Exhibit 14-3. The least squares estimate of b_1 equals
- 1
 - 1
 - 11
 - 11
75. Refer to Exhibit 14-3. The least squares estimate of b_0 equals
- 1
 - 1
 - 11
 - 11

76. Refer to Exhibit 14-3. The sample correlation coefficient equals
- 0.4364
 - 0.4364
 - 0.1905
 - 0.1905
77. Refer to Exhibit 14-3. The coefficient of determination equals
- 0.4364
 - 0.4364
 - 0.1905
 - 0.1905

NARRBEGIN: Exhibit 12-04

Exhibit 14-4

Regression analysis was applied between sales data (Y in \$1,000s) and advertising data (x in \$100s) and the following information was obtained.

$$\hat{Y} = 12 + 1.8x$$

$$n = 17$$

$$SSR = 225$$

$$SSE = 75$$

$$S_{b1} = 0.2683$$

NARREND

78. Refer to Exhibit 14-4. Based on the above estimated regression equation, if advertising is \$3,000, then the point estimate for sales (in dollars) is
- \$66,000
 - \$5,412
 - \$66
 - \$17,400
79. Refer to Exhibit 14-4. The F statistic computed from the above data is
- 3
 - 45
 - 48
 - 50
80. Refer to Exhibit 14-4. To perform an F test, the p -value is
- less than .01
 - between .01 and .025
 - between .025 and .05
 - between .05 and 0.1
81. Refer to Exhibit 14-4. The t statistic for testing the significance of the slope is
- 1.80
 - 1.96
 - 6.708
 - 0.555
82. Refer to Exhibit 14-4. The critical t value for testing the significance of the slope at 95% confidence is
- 1.753
 - 2.131
 - 1.746
 - 2.120

NARRBEGIN: Exhibit 12-05

Exhibit 14-5

The following information regarding a dependent variable (Y) and an independent variable (X) is provided.

Y	X
1	1
2	2
3	3
4	4
5	5

NARREND

83. Refer to Exhibit 14-5. The least squares estimate of the Y intercept is
 - a. 1
 - b. 0
 - c. -1
 - d. 3
84. Refer to Exhibit 14-5. The least squares estimate of the slope is
 - a. 1
 - b. -1
 - c. 0
 - d. 3
85. Refer to Exhibit 14-5. The coefficient of correlation is
 - a. 0
 - b. -1
 - c. 0.5
 - d. 1
86. Refer to Exhibit 14-5. The coefficient of determination is
 - a. 0
 - b. -1
 - c. 0.5
 - d. 1
87. Refer to Exhibit 14-5. The MSE is
 - a. 0
 - b. -1
 - c. 1
 - d. 0.5

NARRBEGIN: Exhibit 12-06

Exhibit 14-6

For the following data the value of SSE = 0.4130.

y Dependent Variable	x Independent Variable
15	4
17	6
23	2
17	4

NARREND

88. Refer to Exhibit 14-6. The slope of the regression equation is
- 18
 - 24
 - 0.707
 - 1.5
89. Refer to Exhibit 14-6. The y intercept is
- 1.5
 - 24
 - 0.50
 - 0.707
90. Refer to Exhibit 14-6. The total sum of squares (SST) equals
- 36
 - 18
 - 9
 - 1296
91. Refer to Exhibit 14-6. The coefficient of determination (r^2) equals
- 0.7071
 - 0.7071
 - 0.5
 - 0.5

NARRBEGIN: Exhibit 12-07

Exhibit 14-7

You are given the following information about y and x.

y	x
Dependent Variable	Independent Variable
5	4
7	6
9	2
11	4

NARREND

92. Refer to Exhibit 14-7. The least squares estimate of b_1 (slope) equals
- 10
 - 10
 - 0.5
 - 0.5
93. Refer to Exhibit 14-7. The least squares estimate of b_0 (intercept) equals
- 10
 - 10
 - 0.5
 - 0.5
94. Refer to Exhibit 14-7. The sample correlation coefficient equals
- 0.3162
 - 0.3162
 - 0.10
 - 0.10

95. Refer to Exhibit 14-7. The coefficient of determination equals
- 0.3162
 - 0.3162
 - 0.10
 - 0.10

NARRBEGIN: Exhibit 12-08

Exhibit 14-8

The following information regarding a dependent variable Y and an independent variable X is provided

$$\begin{array}{ll} \Sigma X = 90 & \Sigma (Y - \bar{Y})(X - \bar{X}) = -156 \\ \Sigma Y = 340 & \Sigma (X - \bar{X})^2 = 234 \\ n = 4 & \Sigma (Y - \bar{Y})^2 = 1974 \\ SSR = 104 & \end{array}$$

NARREND

96. Refer to Exhibit 14-8. The total sum of squares (SST) is
- 156
 - 234
 - 1870
 - 1974
97. Refer to Exhibit 14-8. The sum of squares due to error (SSE) is
- 156
 - 234
 - 1870
 - 1974
98. Refer to Exhibit 14-8. The mean square error (MSE) is
- 1870
 - 13
 - 1974
 - 935
99. Refer to Exhibit 14-8. The slope of the regression equation is
- 0.667
 - 0.667
 - 100
 - 100
100. Refer to Exhibit 14-8. The Y intercept is
- 0.667
 - 0.667
 - 100
 - 100
101. Refer to Exhibit 14-8. The coefficient of correlation is
- 0.2295
 - 0.2295
 - 0.0527
 - 0.0572

NARRBEGIN: Exhibit 12-09

Exhibit 14-9

A regression and correlation analysis resulted in the following information regarding a dependent variable (y) and an independent variable (x).

$$\Sigma X = 90 \qquad \Sigma (Y - \bar{Y})(X - \bar{X}) = 466$$

$$\Sigma Y = 170 \qquad \Sigma (X - \bar{X})^2 = 234$$

$$n = 10 \qquad \Sigma (Y - \bar{Y})^2 = 1434$$

$$SSE = 505.98$$

NARREND

102. Refer to Exhibit 14-9. The least squares estimate of b_1 equals
 - a. 0.923
 - b. 1.991
 - c. -1.991
 - d. -0.923
103. Refer to Exhibit 14-9. The least squares estimate of b_0 equals
 - a. 0.923
 - b. 1.991
 - c. -1.991
 - d. -0.923
104. Refer to Exhibit 14-9. The sum of squares due to regression (SSR) is
 - a. 1434
 - b. 505.98
 - c. 50.598
 - d. 928.02
105. Refer to Exhibit 14-9. The sample correlation coefficient equals
 - a. 0.8045
 - b. -0.8045
 - c. 0
 - d. 1
106. Refer to Exhibit 14-9. The coefficient of determination equals
 - a. 0.6472
 - b. -0.6472
 - c. 0
 - d. 1

NARRBEGIN: Exhibit 12-10

Exhibit 14-10

The following information regarding a dependent variable Y and an independent variable X is provided.

$$\Sigma X = 16 \qquad \Sigma (X - \bar{X})(Y - \bar{Y}) = -8$$

$$\Sigma Y = 28 \qquad \Sigma (X - \bar{X})^2 = 8$$

$$n = 4 \qquad SST = 42$$

$$SSE = 34$$

NARREND

107. Refer to Exhibit 14-10. The slope of the regression function is
- 1
 - 1.0
 - 11
 - 0.0
108. Refer to Exhibit 14-10. The Y intercept is
- 1
 - 1.0
 - 11
 - 0.0
109. Refer to Exhibit 14-10. The coefficient of determination is
- 0.1905
 - 0.1905
 - 0.4364
 - 0.4364
110. Refer to Exhibit 14-10. The coefficient of correlation is
- 0.1905
 - 0.1905
 - 0.4364
 - 0.4364
111. Refer to Exhibit 14-10. The MSE is
- 17
 - 8
 - 34
 - 42
112. Refer to Exhibit 14-10. The point estimate of Y when $X = 3$ is
- 11
 - 14
 - 8
 - 0
113. Refer to Exhibit 14-10. The point estimate of Y when $X = -3$ is
- 11
 - 14
 - 8
 - 0

PROBLEM

1. Assume you have noted the following prices for books and the number of pages that each book contains.

Book	Pages (x)	Price (y)
A	500	\$7.00
B	700	7.50
C	750	9.00
D	590	6.50
E	540	7.50
F	650	7.00
G	480	4.50

- a. Develop a least-squares estimated regression line.
 - b. Compute the coefficient of determination and explain its meaning.
 - c. Compute the correlation coefficient between the price and the number of pages. Test to see if x and y are related. Use $\alpha = 0.10$.
2. Assume you have noted the following prices for books and the number of pages that each book contains.

Book	Pages (x)	Price (y)
A	500	\$7.00
B	700	7.50
C	750	9.00
D	590	6.50
E	540	7.50
F	650	7.00
G	480	4.50

- a. Perform an F test and determine if the price and the number of pages of the books are related. Let $\alpha = 0.01$.
 - b. Perform a t test and determine if the price and the number of pages of the books are related. Let $\alpha = 0.01$.
 - c. Develop a 90% confidence interval for estimating the average price of books that contain 800 pages.
 - d. Develop a 90% confidence interval to estimate the price of a specific book that has 800 pages.
3. The following data represent the number of flash drives sold per day at a local computer shop and their prices.

Price (x)	Units Sold (y)
\$34	3
36	4
32	6
35	5
30	9
38	2
40	1

- a. Develop a least-squares regression line and explain what the slope of the line indicates.
 - b. Compute the coefficient of determination and comment on the strength of relationship between x and y .
 - c. Compute the sample correlation coefficient between the price and the number of flash drives sold. Use $\alpha = 0.01$ to test the relationship between x and y .
4. The following data represent the number of flash drives sold per day at a local computer shop and their prices.

Price (x)	Units Sold (y)
\$34	3
36	4
32	6
35	5
30	9
38	2
40	1

- a. Perform an F test and determine if the price and the number of flash drives sold are related. Let $\alpha = 0.01$.

- b. Perform a t test and determine if the price and the number of flash drives sold are related. Let $\alpha = 0.01$.
5. Shown below is a portion of an Excel output for regression analysis relating Y (dependent variable) and X (independent variable).

ANOVA

	<i>df</i>	<i>SS</i>
Regression	1	110
Residual	8	74
Total	9	184

	<i>Coefficients</i>	<i>Standard Error</i>
Intercept	39.222	5.943
x	-0.5556	0.1611

- What has been the sample size for the above?
 - Perform a t test and determine whether or not X and Y are related. Let $\alpha = 0.05$.
 - Perform an F test and determine whether or not X and Y are related. Let $\alpha = 0.05$.
 - Compute the coefficient of determination.
 - Interpret the meaning of the value of the coefficient of determination that you found in d. Be very specific.
6. Shown below is a portion of a computer output for regression analysis relating Y (dependent variable) and X (independent variable).

ANOVA

	<i>df</i>	<i>SS</i>
Regression	1	24.011
Residual	8	67.989

	<i>Coefficients</i>	<i>Standard Error</i>
Intercept	11.065	2.043
x	-0.511	0.304

- What has been the sample size for the above?
 - Perform a t test and determine whether or not X and Y are related. Let $\alpha = 0.05$.
 - Perform an F test and determine whether or not X and Y are related. Let $\alpha = 0.05$.
 - Compute the coefficient of determination.
 - Interpret the meaning of the value of the coefficient of determination that you found in d. Be very specific.
7. Part of an Excel output relating X (independent variable) and Y (dependent variable) is shown below. Fill in all the blanks marked with "?".

Summary Output

Regression Statistics

Multiple R	0.1347
R Square	?
Adjusted R Square	?
Standard Error	3.3838
Observations	?

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	?	2.7500	?	?	0.632
Residual	?	?	11.45		
Total	14	?			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	8.6	2.2197	?	0.0019
x	0.25	0.5101	?	0.632

8. Shown below is a portion of a computer output for a regression analysis relating Y (dependent variable) and X (independent variable).

ANOVA

	<i>df</i>	<i>SS</i>
Regression	1	115.064
Residual	13	82.936
Total		

	<i>Coefficients</i>	<i>Standard Error</i>
Intercept	15.532	1.457
x	-1.106	0.261

- Perform a t test using the *p*-value approach and determine whether or not Y and X are related. Let $\alpha = 0.05$.
 - Using the *p*-value approach, perform an F test and determine whether or not X and Y are related.
 - Compute the coefficient of determination and fully interpret its meaning. Be very specific.
9. Part of an Excel output relating X (independent variable) and Y (dependent variable) is shown below. Fill in all the blanks marked with "?".

Summary Output

Regression Statistics

Multiple R	?
R Square	0.5149
Adjusted R Square	?
Standard Error	7.3413
Observations	11

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	?	?	?	?	0.0129
Residual	?	?	?		
Total	?	1000			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>
Intercept	?	29.4818	3.7946	0.0043
x	?	0.7000	-3.0911	0.0129

10. Shown below is a portion of a computer output for a regression analysis relating Y (demand) and X (unit price).

ANOVA

	<i>df</i>	<i>SS</i>
Regression	1	5048.818
Residual	46	3132.661
Total	47	8181.479

	<i>Coefficients</i>	<i>Standard Error</i>
Intercept	80.390	3.102
X	-2.137	0.248

- Perform a t test and determine whether or not demand and unit price are related. Let $\alpha = 0.05$.
 - Perform an F test and determine whether or not demand and unit price are related. Let $\alpha = 0.05$.
 - Compute the coefficient of determination and fully interpret its meaning. Be very specific.
 - Compute the coefficient of correlation and explain the relationship between demand and unit price.
11. Shown below is a portion of a computer output for a regression analysis relating supply (Y in thousands of units) and unit price (X in thousands of dollars).

ANOVA

	<i>df</i>	<i>SS</i>
Regression	1	354.689
Residual	39	7035.262

	<i>Coefficients</i>	<i>Standard Error</i>
Intercept	54.076	2.358
X	0.029	0.021

- What has been the sample size for this problem?
 - Perform a t test and determine whether or not supply and unit price are related. Let $\alpha = 0.05$.
 - Perform an F test and determine whether or not supply and unit price are related. Let $\alpha = 0.05$.
 - Compute the coefficient of determination and fully interpret its meaning. Be very specific.
 - Compute the coefficient of correlation and explain the relationship between supply and unit price.
 - Predict the supply (in units) when the unit price is \$50,000.
12. Given below are four observations collected in a regression study on two variables x (independent variable) and y (dependent variable).

x	y
2	4
6	7
9	8
9	9

- Develop the least squares estimated regression equation.
- At 95% confidence, perform a t test and determine whether or not the slope is significantly different from zero.

- c. Perform an F test to determine whether or not the model is significant. Let $\alpha = 0.05$.
- d. Compute the coefficient of determination.

13. Given below are five observations collected in a regression study on two variables, x (independent variable) and y (dependent variable).

x	y
2	4
3	4
4	3
5	2
6	1

- a. Develop the least squares estimated regression equation.
 - b. At 95% confidence, perform a t test and determine whether or not the slope is significantly different from zero.
 - c. Perform an F test to determine whether or not the model is significant. Let $\alpha = 0.05$.
 - d. Compute the coefficient of determination.
 - e. Compute the coefficient of correlation.
14. Below you are given a partial computer output based on a sample of 8 observations, relating an independent variable (x) and a dependent variable (y).

	Coefficient	Standard Error
Intercept	13.251	10.77
X	0.803	0.385

Analysis of Variance

SOURCE	SS
Regression	
Error (Residual)	41.674
Total	71.875

- a. Develop the estimated regression line.
 - b. At $\alpha = 0.05$, test for the significance of the slope.
 - c. At $\alpha = 0.05$, perform an F test.
 - d. Determine the coefficient of determination.
15. Below you are given a partial computer output based on a sample of 8 observations, relating an independent variable (x) and a dependent variable (y).

	Coefficient	Standard Error
Intercept	-9.462	7.032
x	0.769	0.184

Analysis of Variance

SOURCE	SS
Regression	400
Error (Residual)	138

- a. Develop the estimated regression line.
- b. At $\alpha = 0.05$, test for the significance of the slope.
- c. At $\alpha = 0.05$, perform an F test.
- d. Determine the coefficient of determination.

16. The following data represent a company's yearly sales volume and its advertising expenditure over a period of 8 years.

(Y) Sales in Millions of Dollars	(X) Advertising in (\$10,000)
15	32
16	33
18	35
17	34
16	36
19	37
19	39
24	42

- a. Develop a scatter diagram of sales versus advertising and explain what it shows regarding the relationship between sales and advertising.
 - b. Use the method of least squares to compute an estimated regression line between sales and advertising.
 - c. If the company's advertising expenditure is \$400,000, what are the predicted sales? Give the answer in dollars.
 - d. What does the slope of the estimated regression line indicate?
 - e. Compute the coefficient of determination and fully interpret its meaning.
 - f. Use the F test to determine whether or not the regression model is significant at $\alpha = 0.05$.
 - g. Use the t test to determine whether the slope of the regression model is significant at $\alpha = 0.05$.
 - h. Develop a 95% confidence interval for predicting the average sales for the years when \$400,000 was spent on advertising.
 - i. Compute the correlation coefficient.
17. Given below are five observations collected in a regression study on two variables x (independent variable) and y (dependent variable).

x	y
10	7
20	5
30	4
40	2
50	1

- a. Develop the least squares estimated regression equation
- b. At 95% confidence, perform a t test and determine whether or not the slope is significantly different from zero.
- c. Perform an F test to determine whether or not the model is significant. Let $\alpha = 0.05$.
- d. Compute the coefficient of determination.
- e. Compute the coefficient of correlation.

18. Below you are given a partial computer output based on a sample of 14 observations, relating an independent variable (x) and a dependent variable (y).

Predictor	Coefficient	Standard Error
Constant	6.428	1.202
X	0.470	0.035

Analysis of Variance

SOURCE	SS
Regression	958.584
Error (Residual)	
Total	1021.429

- Develop the estimated regression line.
 - At $\alpha = 0.05$, test for the significance of the slope.
 - At $\alpha = 0.05$, perform an F test.
 - Determine the coefficient of determination.
 - Determine the coefficient of correlation.
19. Below you are given a partial computer output based on a sample of 21 observations, relating an independent variable (x) and a dependent variable (y).

Predictor	Coefficient	Standard Error
Constant	30.139	1.181
X	-0.252	0.022

Analysis of Variance

SOURCE	SS
Regression	1,759.481
Error	259.186

- Develop the estimated regression line.
 - At $\alpha = 0.05$, test for the significance of the slope.
 - At $\alpha = 0.05$, perform an F test.
 - Determine the coefficient of determination.
 - Determine the coefficient of correlation.
20. An automobile dealer wants to see if there is a relationship between monthly sales and the interest rate. A random sample of 4 months was taken. The results of the sample are presented below. The estimated least squares regression equation is

$$\hat{Y} = 75.061 - 6.254X$$

Y	X
Monthly Sales	Interest Rate (In Percent)
22	9.2
20	7.6
10	10.4
45	5.3

- a. Obtain a measure of how well the estimated regression line fits the data.
 - b. You want to test to see if there is a significant relationship between the interest rate and monthly sales at the 1% level of significance. State the null and alternative hypotheses.
 - c. At 99% confidence, test the hypotheses.
 - d. Construct a 99% confidence interval for the average monthly sales for all months with a 10% interest rate.
 - e. Construct a 99% confidence interval for the monthly sales of one month with a 10% interest rate.
21. Jason believes that the sales of coffee at his coffee shop depend upon the weather. He has taken a sample of 6 days. Below you are given the results of the sample.

Cups of Coffee Sold	Temperature
350	50
200	60
210	70
100	80
60	90
40	100

- a. Which variable is the dependent variable?
 - b. Compute the least squares estimated line.
 - c. Compute the correlation coefficient between temperature and the sales of coffee.
 - d. Is there a significant relationship between the sales of coffee and temperature? Use a .05 level of significance. Be sure to state the null and alternative hypotheses.
 - e. Predict sales of a 90 degree day.
22. Researchers have collected data on the hours of television watched in a day and the age of a person. You are given the data below.

Hours of Television	Age
1	45
3	30
4	22
3	25
6	5

- a. Determine which variable is the dependent variable.
 - b. Compute the least squares estimated line.
 - c. Is there a significant relationship between the two variables? Use a .05 level of significance. Be sure to state the null and alternative hypotheses.
 - d. Compute the coefficient of determination. How would you interpret this value?
23. Given below are seven observations collected in a regression study on two variables, X (independent variable) and Y (dependent variable).

X	Y
2	12
3	9
6	8
7	7
8	6
7	5
9	2

- Develop the least squares estimated regression equation.
- At 95% confidence, perform a t test and determine whether or not the slope is significantly different from zero.
- Perform an F test to determine whether or not the model is significant. Let $\alpha = 0.05$.
- Compute the coefficient of determination.

24. The owner of a retail store randomly selected the following weekly data on profits and advertising cost.

Week	Advertising Cost (\$)	Profit (\$)
1	0	200
2	50	270
3	250	420
4	150	300
5	125	325

- Write down the appropriate linear relationship between advertising cost and profits. Which is the dependent variable? Which is the independent variable?
 - Calculate the least squares estimated regression line.
 - Predict the profits for a week when \$200 is spent on advertising.
 - At 95% confidence, test to determine if the relationship between advertising costs and profits is statistically significant.
 - Calculate the coefficient of determination.
25. The owner of a bakery wants to analyze the relationship between the expenditure of a customer and the customer's income. A sample of 5 customers is taken and the following information was obtained.

Y Expenditure	X Income (In Thousands)
.45	20
10.75	19
5.40	22
7.80	25
5.60	14

The least squares estimated line is $\hat{Y} = 4.348 + 0.0826 X$.

- Obtain a measure of how well the estimated regression line fits the data.
 - You want to test to see if there is a significant relationship between expenditure and income at the 5% level of significance. Be sure to state the null and alternative hypotheses.
 - Construct a 95% confidence interval estimate for the average expenditure for all customers with an income of \$20,000.
 - Construct a 95% confidence interval estimate for the expenditure of one customer whose income is \$20,000.
26. Below you are given information on annual income and years of college education.
- | Income (In Thousands) | Years of College |
|-----------------------|------------------|
| 28 | 0 |
| 40 | 3 |
| 36 | 2 |
| 28 | 1 |
| 48 | 4 |
- Develop the least squares regression equation.
 - Estimate the yearly income of an individual with 6 years of college education.

- c. Compute the coefficient of determination.
- d. Use a t test to determine whether the slope is significantly different from zero. Let $\alpha = 0.05$.
- e. At 95% confidence, perform an F test and determine whether or not the model is significant.

27. Below you are given information on a woman's age and her annual expenditure on purchase of books.

Age	Annual Expenditure (\$)
18	210
22	180
21	220
28	280

- a. Develop the least squares regression equation.
- b. Compute the coefficient of determination.
- c. Use a t test to determine whether the slope is significantly different from zero. Let $\alpha = 0.05$.
- d. At 95% confidence, perform an F test and determine whether or not the model is significant.

28. The following sample data contains the number of years of college and the current annual salary for a random sample of heavy equipment salespeople.

Years of College	Annual Income (In Thousands)
2	20
2	23
3	25
4	26
3	28
1	29
4	27
3	30
4	33
4	35

- a. Which variable is the dependent variable? Which is the independent variable?
- b. Determine the least squares estimated regression line.
- c. Predict the annual income of a salesperson with one year of college.
- d. Test if the relationship between years of college and income is statistically significant at the .05 level of significance.
- e. Calculate the coefficient of determination.
- f. Calculate the sample correlation coefficient between income and years of college. Interpret the value you obtain.

29. The following data shows the yearly income (in \$1,000) and age of a sample of seven individuals.

Income (in \$1,000)	Age
20	18
24	20
24	23
25	34
26	24
27	27
34	27

- a. Develop the least squares regression equation.
- b. Estimate the yearly income of a 30-year-old individual.
- c. Compute the coefficient of determination.
- d. Use a t test to determine whether the slope is significantly different from zero. Let $\alpha = 0.05$.
- e. At 95% confidence, perform an F test and determine whether or not the model is significant.

30. The following data show the results of an aptitude test (Y) and the grade point average of 10 students.

Aptitude Test Score (Y)	GPA (X)
26	1.8
31	2.3
28	2.6
30	2.4
34	2.8
38	3.0
41	3.4
44	3.2
40	3.6
43	3.8

- Develop a least squares estimated regression line.
 - Compute the coefficient of determination and comment on the strength of the regression relationship.
 - Is the slope significant? Use a t test and let $\alpha = 0.05$.
 - At 95% confidence, test to determine if the model is significant (i.e., perform an F test).
31. Shown below is a portion of the computer output for a regression analysis relating sales (Y in millions of dollars) and advertising expenditure (X in thousands of dollars).

Predictor	Coefficient	Standard Error
Constant	4.00	0.800
X	0.12	0.045

Analysis of Variance

SOURCE	DF	SS
Regression	1	1,400
Error	18	3,600

- What has been the sample size for the above?
 - Perform a t test and determine whether or not advertising and sales are related. Let $\alpha = 0.05$.
 - Compute the coefficient of determination.
 - Interpret the meaning of the value of the coefficient of determination that you found in Part c. Be very specific.
 - Use the estimated regression equation and predict sales for an advertising expenditure of \$4,000. Give your answer in dollars.
32. A company has recorded data on the daily demand for its product (Y in thousands of units) and the unit price (X in hundreds of dollars). A sample of 15 days demand and associated prices resulted in the following data.

$$\begin{aligned}\Sigma X &= 75 & \Sigma (Y - \bar{Y})(X - \bar{X}) &= -59 \\ \Sigma Y &= 135 & \Sigma (X - \bar{X})^2 &= 94 \\ \Sigma (Y - \bar{Y})^2 &= 100 \\ \text{SSE} &= 62.9681\end{aligned}$$

- Using the above information, develop the least-squares estimated regression line and write the equation.
- Compute the coefficient of determination.

- c. Perform an F test and determine whether or not there is a significant relationship between demand and unit price. Let $\alpha = 0.05$.
 - d. Would the demand ever reach zero? If yes, at what price would the demand be zero?
33. A regression and correlation analysis resulted in the following information regarding an independent variable (x) and a dependent variable (y).

$$\begin{array}{ll} \Sigma X = 42 & \Sigma (Y - \bar{Y})(X - \bar{X}) = 37 \\ \Sigma Y = 63 & \Sigma (X - \bar{X})^2 = 84 \\ n = 7 & \Sigma (Y - \bar{Y})^2 = 28 \\ & \Sigma \{Y - \hat{Y}\}^2 = 11.7024 \end{array}$$

- a. Develop the least squares estimated regression equation.
- b. At 95% confidence, perform a t test and determine whether or not the slope is significantly different from zero.
- c. Perform an F test to determine whether or not the model is significant. Let $\alpha = 0.05$.
- d. Compute the coefficient of determination.